

IBM HR Analytics Employee Attrition

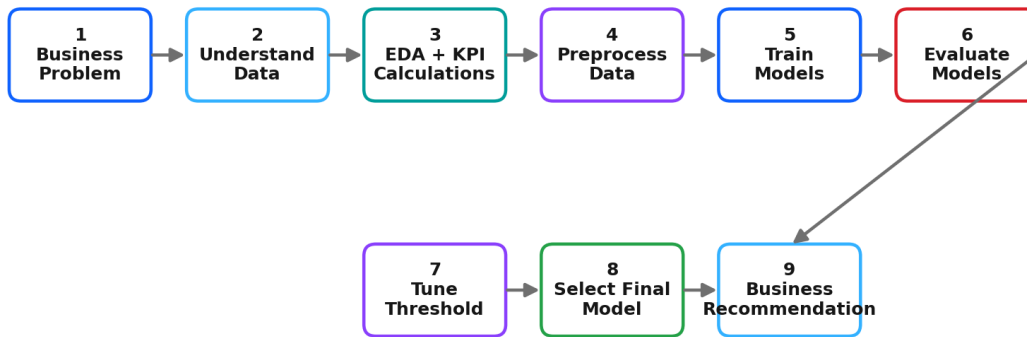
Foundation Manual: Concepts, Calculations, Charts, and Models

Prepared for Ruturaj Mokashi

A practical guide to understand everything we did in the attrition project - from groupby KPIs to model evaluation.

IBM HR Attrition Project - Learning Workflow

The full journey from business question to model decision



Key idea: We did not just run code. We followed a business problem -> data -> insight -> model -> decision process.
Final choice: Random Forest with threshold 0.25 because it gave the strongest recall and F1-score for catching leavers.

Figure 1. Project workflow map.

How to use this manual

Read it as a concept book, not as a code notebook. Each technical idea is explained with plain language, then connected back to your IBM HR attrition project.

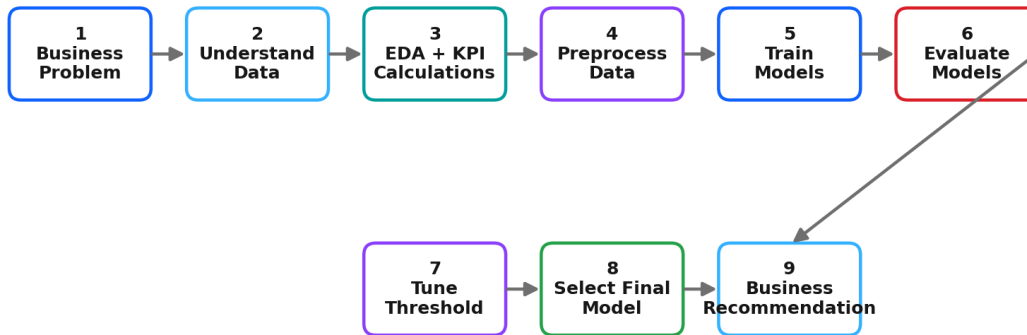
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1. Big Picture: What We Built and Why

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The full journey from business question to model decision



Key idea: We did not just run code. We followed a business problem -> data -> insight -> model -> decision process.

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Figure 2. Clean workflow view of the whole project.

The project in one simple story

We started with a business problem: employees leaving the company can be expensive, disruptive, and difficult to manage. The goal was not simply to make charts or train models. The goal was to understand which employees are more likely to leave and build a practical early-warning model for HR.

The project moved through four layers. First, we understood the dataset. Second, we used EDA to calculate attrition patterns across groups. Third, we prepared the data so machine learning models could use it. Fourth, we trained, evaluated, compared, and selected the model that best matched the HR objective.

Main learning point

A professional project is not model-first. It is problem-first. The model is only useful if it supports the business decision.

What final model did we select?

The final selected model was Random Forest with a prediction threshold of 0.25. This was selected because it achieved the best recall and F1-score for finding employees who actually left.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC	Meaning
LogReg + MinMaxScaler	76.4%	37.4%	69.0%	48.5%	82.2%	59.4%	Best probability ranking
Random Forest Default	83.0%	42.3%	15.5%	22.7%	77.6%	42.7%	High accuracy but misses leavers
Random Forest threshold 0.25	80.0%	42.7%	70.4%	53.2%	77.6%	42.7%	Best early-warning balance
XGBoost	82.1%	44.4%	45.1%	44.8%	76.3%	47.0%	Good precision but lower recall

Why not choose highest accuracy?

Because the default Random Forest had high accuracy but recall of only 15.5%. It was good at predicting stayers but bad at catching leavers. For HR attrition, missing real leavers is a serious problem.

2. Dataset and Business Objective

Dataset structure

The IBM HR Analytics dataset has one row per employee. Each row contains employee characteristics, such as age, job role, income, overtime status, satisfaction scores, and years at company. The target column is Attrition.

Concept	Project meaning
Row	One employee
Column / feature	One piece of information about an employee, such as Age or JobRole
Target	Attrition: the answer we want to predict
Attrition = 0	Employee stayed
Attrition = 1	Employee left

Dataset numbers

Employee Status	Count	Percentage
Stayed	1,233	83.88%
Left	237	16.12%
Total	1,470	100.00%

Why imbalance matters

Only 16.12% of employees left. This means a lazy model could predict Stayed often and still look accurate. That is why we focused on recall, F1-score, PR-AUC, and business meaning - not just accuracy.

Business objective

The business objective was to identify employees who may be at higher risk of leaving. The model should support proactive HR action, such as workload review, career development conversations, manager check-ins, and retention support.

Ethical reminder

A model prediction is not a judgment about a person. It should support thoughtful HR conversations, not create automatic negative decisions about employees.

3. KPIs, groupby, Aggregation, and Rate Calculations

What is a KPI?

A KPI is a Key Performance Indicator. In this project, the most important KPI was the attrition rate. It tells us what percentage of employees left in a group.

$$\text{Attrition Rate (\%)} = \text{Number of employees who left} / \text{Total number of employees} \times 100$$

KPI Calculation: From Employee Rows to Attrition Rate

Most EDA charts were built by grouping employees, counting leavers, and calculating rates.



Formula used in many charts:

Attrition Rate (%) = Number of employees who left in that group / Total employees in that group × 100

Example: Overtime Yes = 127 leavers / 416 employees × 100 = 30.5%

Figure 3. How groupby and aggregation created attrition KPIs.

What is groupby?

groupby is like putting employees into buckets. For example, groupby("OverTime") creates two buckets: employees with OverTime = Yes and employees with OverTime = No. Then we count how many people are inside each bucket and how many left.

Step	What happens	Project example
1. Choose group	Select the column to make buckets	OverTime
2. Count total	Count employees in each bucket	416 overtime employees, 1,054 non-overtime employees
3. Count leavers	Add Attrition values because Left = 1 and Stayed = 0	127 overtime leavers, 110 non-overtime leavers
4. Calculate rate	Leavers / Total × 100	127 / 416 × 100 = 30.5%

Why summing Attrition works

We converted Attrition to numbers: Stayed = 0 and Left = 1. This makes calculation easy. If a group has values [1, 0, 0, 1, 1], the sum is 3, meaning 3 employees left.

```
summary = df.groupby("OverTime")["Attrition"].agg(["sum", "count", "mean"])
# sum = number of leavers
# count = total employees
# mean = attrition rate as decimal
```

Did we use joins?

No join was needed in this project because the dataset was already one complete table. A join is needed when useful information is split across multiple tables, such as employee table + department table + salary table. Here, all needed columns were already in one file.

Core EDA formula

Most charts were built from the same idea: group employees -> count total -> count leavers -> calculate attrition rate -> visualize the result.

4. Chart Concepts: How to Read the Visuals

Bar chart

A bar chart compares values across categories. In our project, bar charts often showed attrition rate by group. The height or length of the bar represented the attrition rate.

Dot plot

A dot plot is useful when we want to compare a few category rates without heavy bars. The dot position shows the value. It is cleaner than a bar chart when there are few groups.

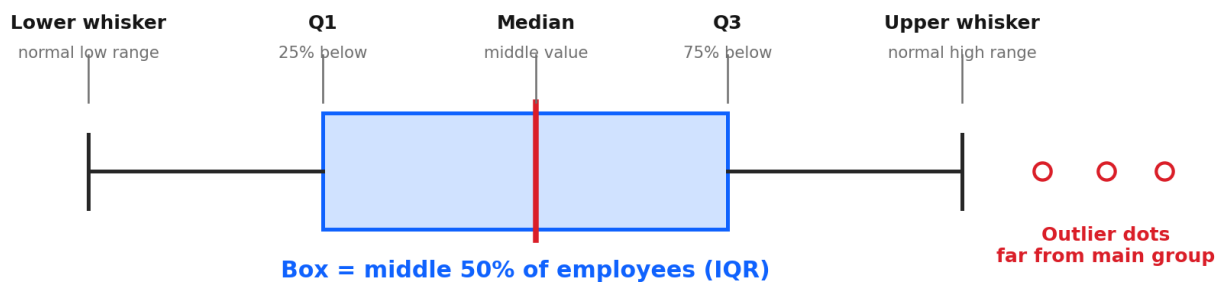
Line chart

A line chart is useful when the x-axis has an order, such as age bands, years at company, satisfaction score, or threshold values. The line helps show trend or movement.

Box plot and violin plot

Box Plot Anatomy: What the Box and Whiskers Mean

A box plot compresses many individual values into five visual landmarks.



Reading rule: compare medians first; then compare box width, whiskers, and outlier dots.

Figure 4. Box plot anatomy explained visually.

A box plot looks mysterious at first because it compresses many individual data points into a small shape. The box is the middle 50% of the data. The line inside the box is the median. The whiskers show the normal low and high range. Dots beyond the whiskers are possible outliers.

Box plot part	Simple meaning
Median	The middle value. Half the employees are below it, half are above it.
Q1	25% of values are below this point.
Q3	75% of values are below this point.
IQR	$Q3 - Q1$. This is the middle 50% spread.
Whiskers	Normal low/high values, usually up to $1.5 \times IQR$ from the box.
Outlier dots	Values far away from the main group.

Why the income chart was useful

The monthly income violin/box chart compared the income distribution of employees who stayed and employees who left. It showed that the median income of leavers was lower. The violin shape showed where employees were concentrated, while the box showed the middle 50%.

Heatmap

A heatmap uses color intensity to show value size. Darker color usually means higher value. We initially used a heatmap for job satisfaction, but the cleaner line chart was easier to understand.

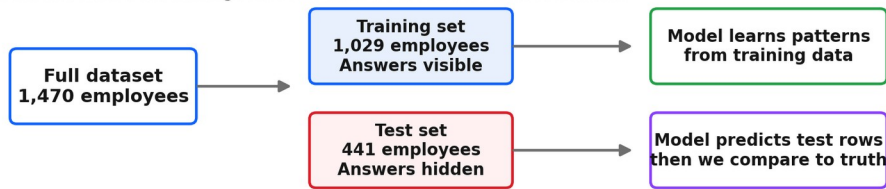
Confusion matrix, ROC curve, PR curve

These are evaluation charts, not EDA charts. They do not describe the raw company data. They describe how the trained model performed on the test set. They are explained in detail in the evaluation section.

5. Preprocessing: Splitting, Scaling, Encoding, and Pipelines

Train-Test Split: Why it is still prediction

The model learns from training rows. Test answers are hidden until evaluation.



School analogy: Training set = homework with answers. Test set = exam. The teacher has the answers, but the student does not see them while answering.
Project meaning: Attrition data exists, but prediction is honest because the model predicts the test rows before seeing their Attrition values.

Figure 5. Train-test split explained.

Why train-test split matters

The model learned from 1,029 employees and was evaluated on 441 employees. The test answers were hidden from the model until after prediction. This is why it is still called prediction even though the dataset already exists.

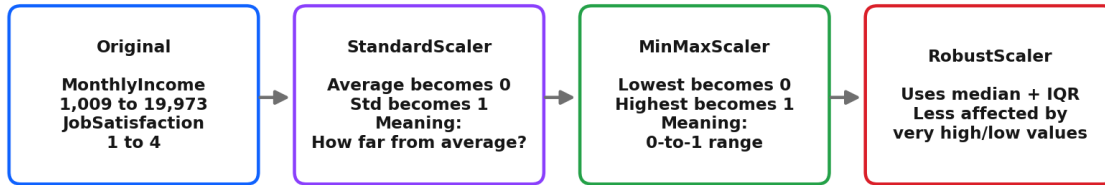
Numeric and categorical features

Feature type	Examples	What we did
Numeric	Age, MonthlyIncome, YearsAtCompany	Impute missing values and scale for Logistic Regression
Categorical	JobRole, Department, OverTime	Impute missing values and encode into numbers

Scaling explained deeply

Scaling: Same Meaning, Different Number Shape

Scaling does not change business meaning. It changes the numeric language for the model.



Project result: MinMaxScaler gave the best Logistic Regression performance: recall 69.0%, F1-score 48.5%, ROC-AUC 82.2%.

Figure 6. Scaling transforms the numeric language used by the model.

Scaling does not change business meaning. It only changes how numbers are represented. MonthlyIncome is still income, and JobSatisfaction is still satisfaction. But the numeric scale becomes easier for some models to handle.

Scaler	What it does	Simple analogy	When useful
StandardScaler	Centers values around 0 and scales by standard deviation	How far from the average am I?	Common default for Logistic Regression, SVM, PCA
MinMaxScaler	Transforms values to a 0-to-1 range	Where am I between minimum and maximum?	Useful when a bounded scale helps
RobustScaler	Uses median and IQR instead of mean/std	Focus on the middle group, ignore extreme values more	Useful with strong outliers

Source reference: [1, 2, 3]

Our scaler result

For Logistic Regression, MinMaxScaler performed slightly better than StandardScaler and RobustScaler. This does not mean MinMaxScaler is always best. It means it worked best for this dataset and this model.

Encoding explained

Machine learning models need numbers. They cannot directly understand text such as Sales, Travel_Rarely, or Yes. Encoding converts categorical text into numeric representation.

OneHotEncoder

One-hot encoding creates a new 0/1 column for each category. For example, OverTime becomes OverTime_No and OverTime_Yes. This avoids pretending that categories have an order.

Original OverTime	OverTime_No	OverTime_Yes
Yes	0	1
No	1	0
Yes	0	1

TargetEncoder

Target encoding replaces categories with target-based averages. For example, if Sales Representative has high attrition rate in training data, it gets a higher encoded value. It can be powerful, but it uses the target column, so it can cause data leakage if fitted incorrectly.

Data leakage danger

TargetEncoder must be fitted only on training data. If it sees test target values, the model has indirectly seen exam answers.

Source reference: [10]

Pipeline and ColumnTransformer

A pipeline keeps preprocessing and modeling together. ColumnTransformer applies different transformations to different column types, such as scaling numeric columns and one-hot encoding categorical columns. This makes the workflow reusable and reduces mistakes.

Source reference: [5, 6]

Pipeline = preprocessing steps + model

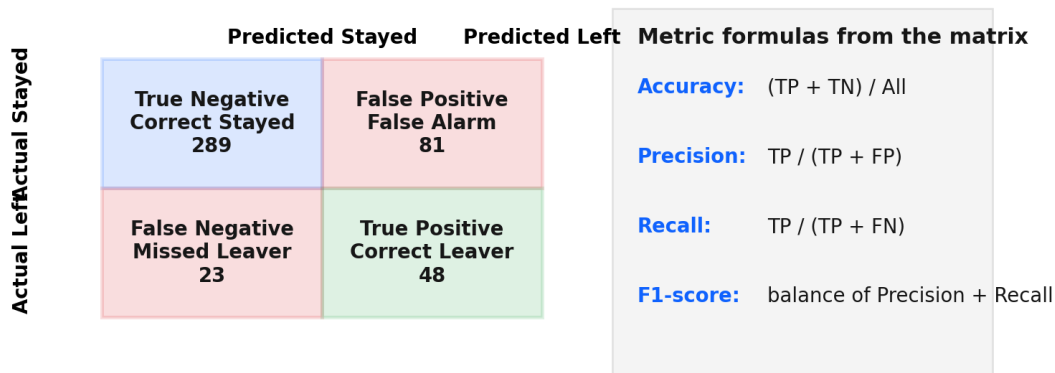
ColumnTransformer = numeric pipeline for numeric columns + categorical pipeline for categorical columns

6. Model Evaluation: The Part That Felt Like Technical Gibberish

This section is the foundation. Evaluation metrics are just different ways of asking: what mistakes did the model make, and which mistakes matter most for the business?

Confusion Matrix: The Four Possible Prediction Outcomes

It compares what really happened with what the model predicted.



In HR attrition, the most painful error is often the False Negative: a real leaver the model failed to flag.

Figure 7. Confusion matrix anatomy using our project values.

Confusion matrix step-by-step

The confusion matrix is built by comparing two columns for every test employee: the real answer and the model prediction.

Actual answer	Model prediction	Outcome name	Meaning
Stayed	Stayed	True Negative	Correctly predicted stayed
Stayed	Left	False Positive	False alarm
Left	Stayed	False Negative	Missed real leaver
Left	Left	True Positive	Correctly caught leaver

Our baseline confusion matrix

	Predicted Stayed	Predicted Left
Actual Stayed	289	81
Actual Left	23	48

So the baseline model correctly caught 48 leavers and missed 23 leavers. That is why recall was $48 / (48 + 23) = 67.6\%$.

Metrics: what each one really asks

Evaluation Metrics: Which Question Does Each Metric Answer?

Accuracy

Out of all employees, how many predictions were correct?
Useful, but dangerous with imbalanced data.

Precision

When the model says "will leave", how often is it right?
Controls false alarms.

Recall

Of all real leavers, how many did we catch?
Critical for HR early-warning.

F1-score

Is the precision-recall trade-off balanced?
Good single threshold metric.

ROC-AUC

Can the model rank leavers above stayers overall?
Probability-ranking metric.

PR-AUC

How good is the model for the rare class across thresholds?
Important for imbalanced data.

Figure 8. Metric meaning map.

Metric	Formula	Plain English	Why it matters here
Accuracy	$(TP + TN) / All$	Overall correct predictions	Can be misleading because most employees stayed
Precision	$TP / (TP + FP)$	When model says Left, how often is it right?	Measures false alarms
Recall	$TP / (TP + FN)$	Of all real leavers, how many did we catch?	Very important for HR early warning
F1-score	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$	Balance between precision and recall	Good for comparing threshold-based models
ROC-AUC	Area under ROC curve	How well probabilities rank leavers above stayers	Good overall ranking view
PR-AUC	Area under precision-recall curve	How well model performs for the rare positive class	Useful for imbalanced data

Source reference: [4]

Why accuracy misled us

The default Random Forest had 83.0% accuracy. That looked good. But its recall was only 15.5%, meaning it missed most leavers. This happened because most employees stayed, so predicting Stayed often can create high accuracy without solving the HR problem.

Metric priority for this project

For employee attrition, recall and F1-score were more important than accuracy because HR wants to catch employees at risk before they leave.

ROC-AUC and PR-AUC without mystery

ROC-AUC and PR-AUC use predicted probabilities, not just final 0/1 predictions. They judge how well the model ranks employees from lower risk to higher risk.

ROC-AUC asks: if I randomly pick one leaver and one stayer, how often does the model give the leaver a higher risk score? PR-AUC focuses more directly on the positive class, which is employees who left. That is why PR-AUC is especially helpful when leavers are rare.

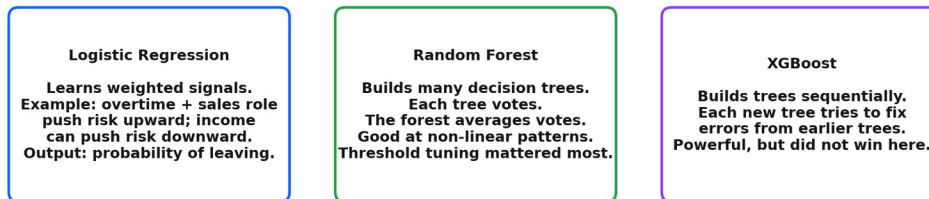
Threshold vs ranking

Changing the threshold changes accuracy, precision, recall, and F1-score. It does not change ROC-AUC or PR-AUC because those are based on probability ranking, not the final threshold label.

7. Models: What They Do and Why We Used Them

Model Intuition: What Each Model is Doing

The models differ in how they learn patterns from employee data.



Project lesson: A trained model does not automatically mean best model. We selected the model that best matched the business objective: catching leavers.

Figure 9. Model intuition overview.

Logistic Regression

Logistic Regression is a classification model. Despite the word regression, it predicts classes, such as Left or Stayed. It calculates a risk score by giving weights to features. Positive weights push the probability toward leaving. Negative weights push it toward staying.

We chose it first because it is simple, strong as a baseline, and interpretable through coefficients.

Source reference: [7]

How it works in our project

If an employee works overtime, is a Sales Representative, travels frequently, and has short tenure, those signals can push the predicted probability of leaving upward. If the employee has higher income, no overtime, and more working years, those signals can push the probability downward.

Random Forest

Random Forest builds many decision trees. Each tree asks simple yes/no questions about the employee. For example: Is OverTime = Yes? Is MonthlyIncome low? Is YearsAtCompany <= 1? The forest combines many trees so that one tree does not dominate.

The default Random Forest was too conservative. It predicted too few leavers. After lowering the threshold to 0.25, it became the best model for the HR objective.

Source reference: [8]

XGBoost

XGBoost is a boosted tree model. Unlike Random Forest, which builds many trees independently, boosting builds trees sequentially. Each new tree tries to fix mistakes from earlier trees. XGBoost is very popular in industry and competitions, but in this project it did not beat the tuned Random Forest.

Source reference: [9]

Industry model overview

Model family	Common models	Typical business use
Linear models	Linear Regression, Logistic Regression	Baseline models, interpretable scoring, risk prediction
Tree models	Decision Tree, Random Forest	Customer churn, credit risk, HR risk, tabular datasets
Boosted trees	XGBoost, LightGBM, CatBoost	High-performance tabular prediction, fraud, pricing, churn
Neural networks	MLP, CNN, Transformer models	Images, text, speech, complex unstructured data
Time-series models	ARIMA, Prophet, LSTM, Temporal Fusion Transformer	Forecasting sales, demand, inventory, energy
LLMs / foundation models	GPT-style models, embeddings, RAG systems	Text generation, semantic search, document Q&A, assistants

Professional lesson

Modern does not always mean better. For tabular business data, Logistic Regression, Random Forest, XGBoost, LightGBM, and CatBoost are still very common and very useful.

8. Threshold Tuning and Final Model Selection

What is a threshold?

Most classification models first produce a probability. Example: this employee has 0.31 predicted probability of leaving. Then we need a threshold to convert that probability into a final label.

Probability of leaving	Threshold	Final prediction
0.31	0.50	Stayed, because $0.31 < 0.50$
0.31	0.25	Left, because $0.31 \geq 0.25$

The default threshold is usually 0.50, but that is not always best. For HR attrition, we may prefer a lower threshold because we want to catch more potential leavers.

Why threshold 0.25 worked

At threshold 0.50, Random Forest recall was only 15.5%. At threshold 0.25, recall improved to 70.4% and F1-score improved to 53.2%. That made the Random Forest useful as an early-warning model.

Random Forest Threshold Tuning

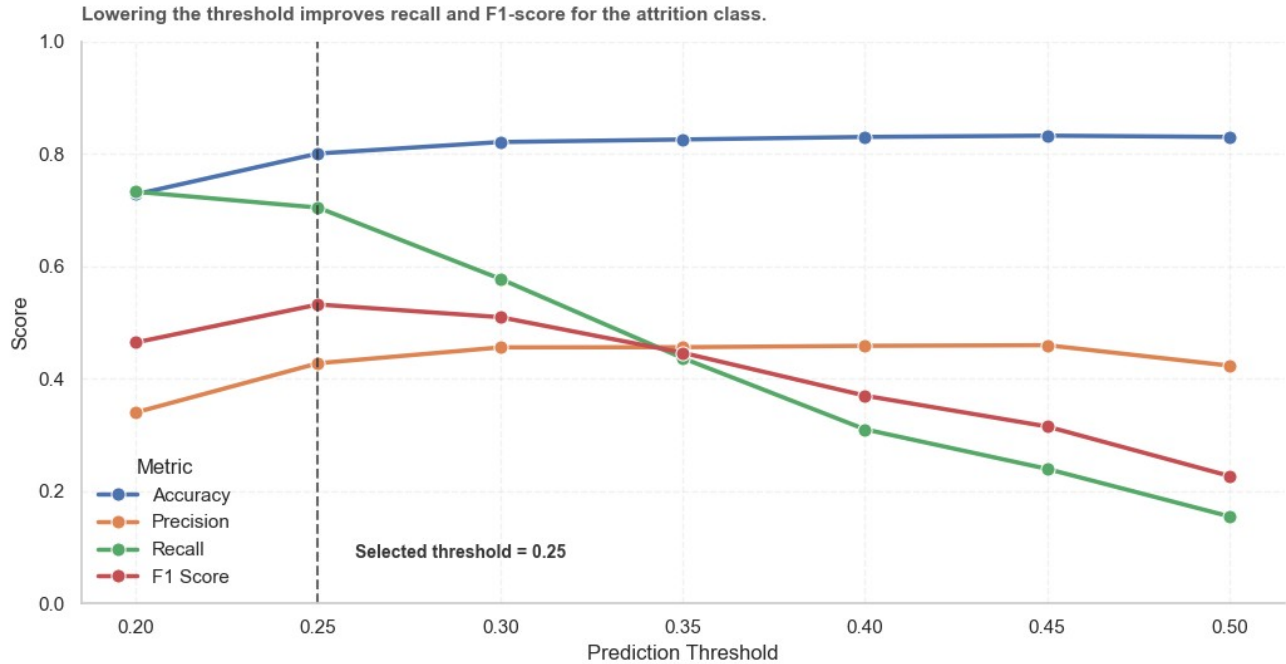


Figure 10. Threshold tuning showed that 0.25 gave a better precision-recall balance.

Final model decision logic

Question	Answer
Do we only want highest accuracy?	No. Accuracy can hide missed leavers.
Do we want to catch many real leavers?	Yes. Recall is important.
Do we also want reasonable false alarms?	Yes. F1-score balances precision and recall.
Which model best matches that?	Random Forest with threshold 0.25.

Final selected model

Random Forest with threshold 0.25 was selected because it caught the most real leavers and had the best F1-score among candidate models.

9. Chart-by-Chart Project Explanation

This section explains what each important chart answered, how the numbers were calculated, and how to read it. The charts are not decoration. Each chart answers one business question.

1. Employee Attrition Overview

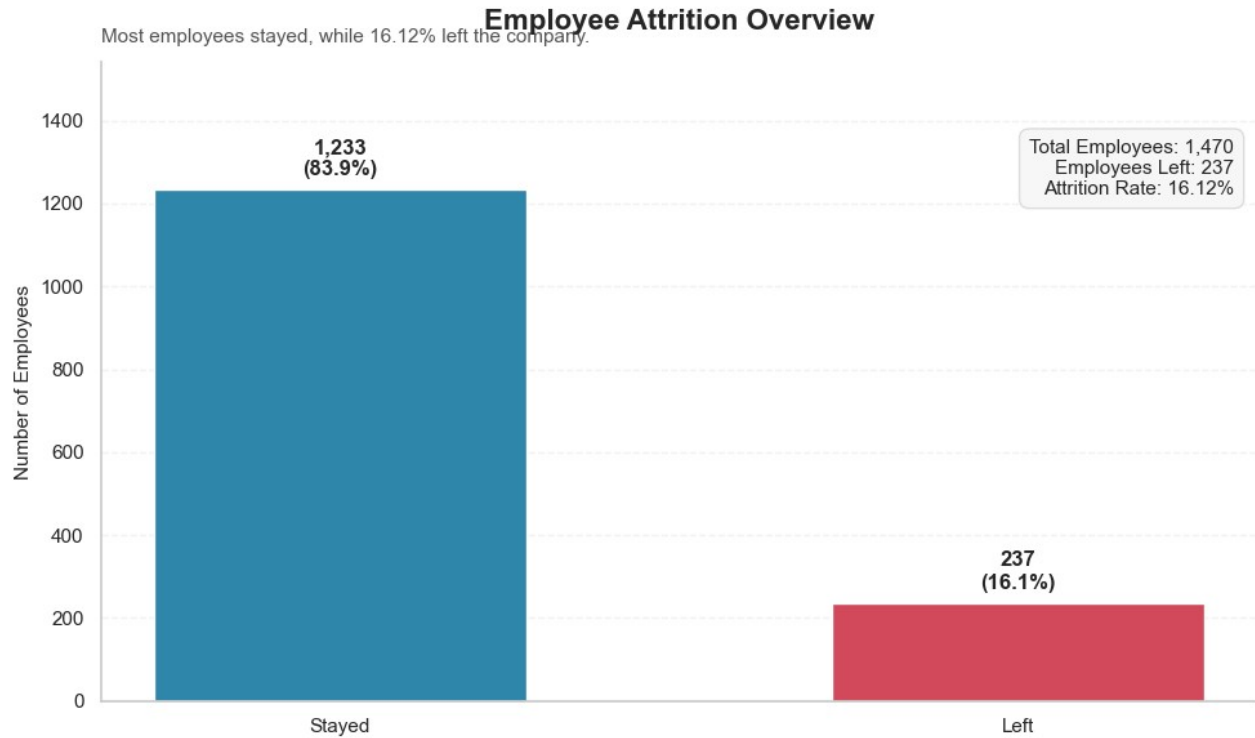


Figure 11. Employee Attrition Overview.

Question answered	How many employees stayed vs left?
Calculation / method	value_counts on Attrition; attrition rate = $237 / 1470 \times 100$.
How to read it	Shows the dataset is imbalanced: 16.12% left, 83.88% stayed.

2. Attrition Rate by Overtime

Attrition Rate by Overtime Status

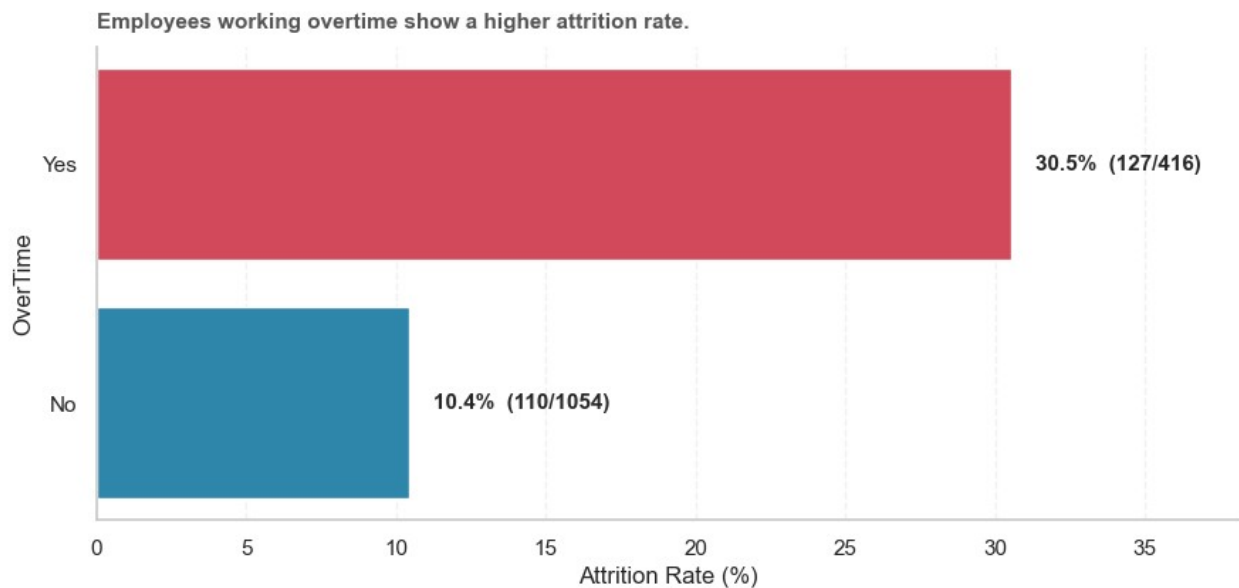


Figure 12. Attrition Rate by Overtime.

Question answered	Do overtime employees leave more?
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Calculation / method	groupby OverTime; count total and leavers; leavers / total × 100.
How to read it	Overtime Yes has much higher attrition: 30.5% vs 10.4%.

3. Attrition Rate by Business Travel

Attrition Rate by Business Travel Frequency

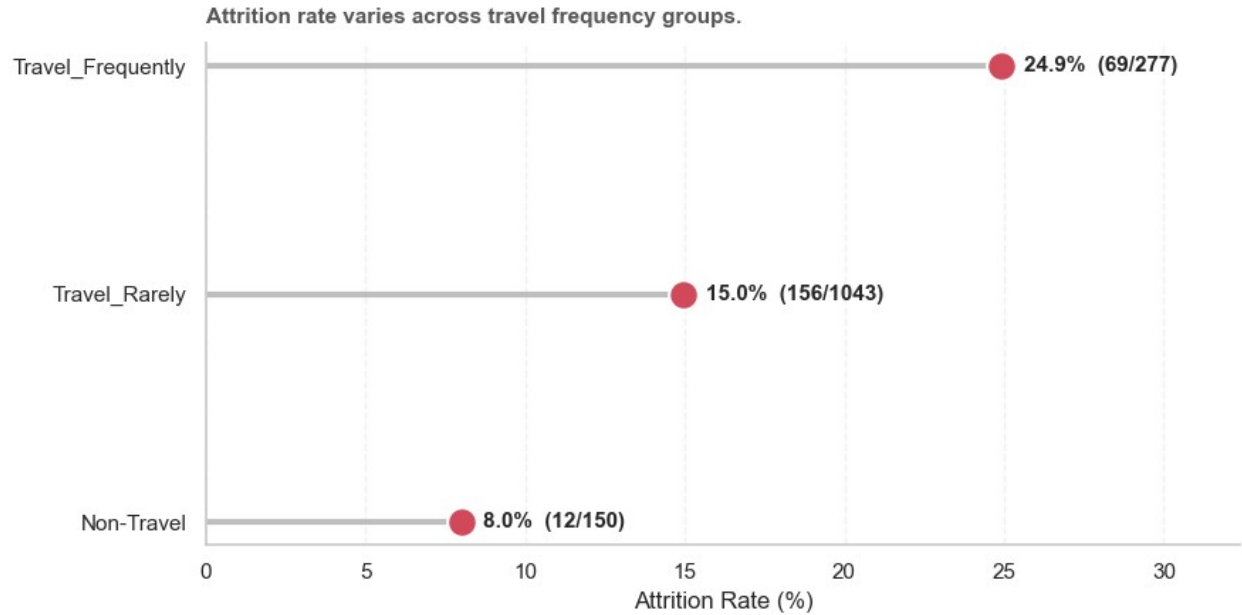


Figure 13. Attrition Rate by Business Travel.

Question answered	Does travel frequency matter?
Calculation / method	groupby BusinessTravel; calculate attrition rate per travel group.
How to read it	Travel_Frequently has the highest attrition rate.

4. Attrition Rate by Job Role

Attrition Rate by Job Role

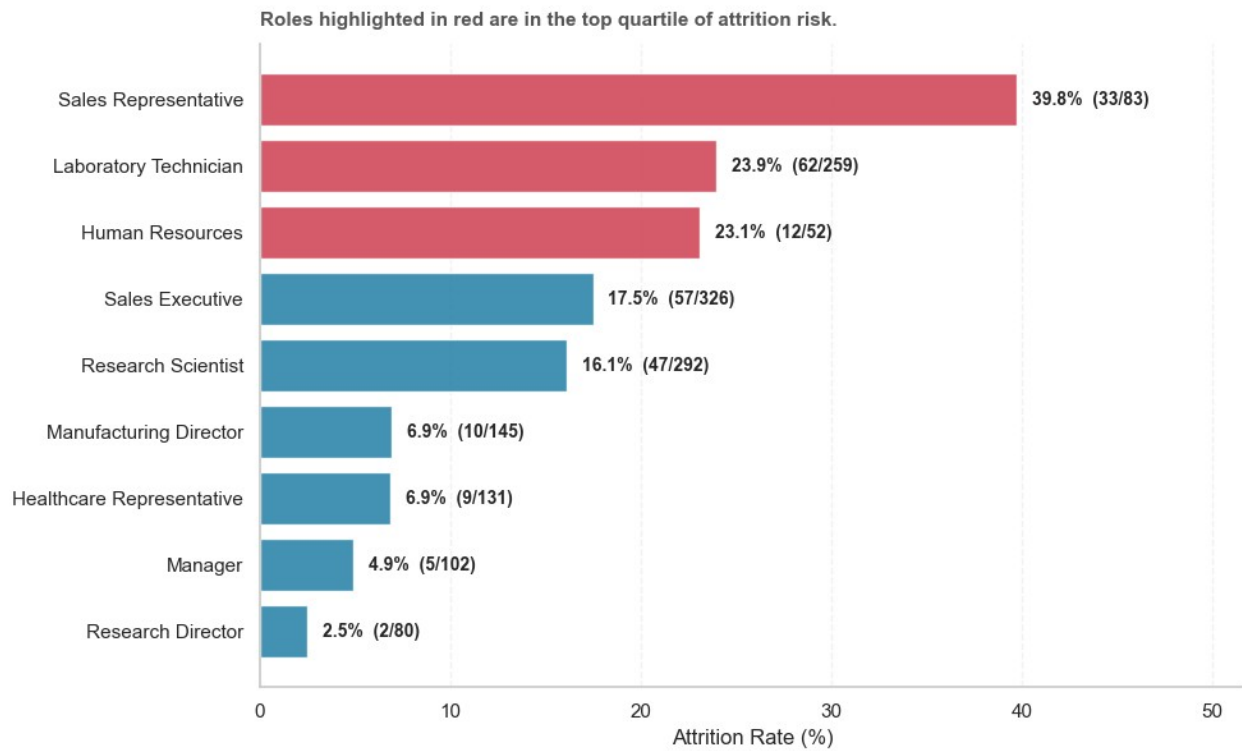


Figure 14. Attrition Rate by Job Role.

Question answered	Which roles have the highest attrition?
Calculation / method	groupby JobRole; calculate attrition rate; sort descending.
How to read it	Sales Representative is the highest-risk role.

5. Monthly Income Distribution by Attrition Status

Monthly Income Distribution by Attrition Status

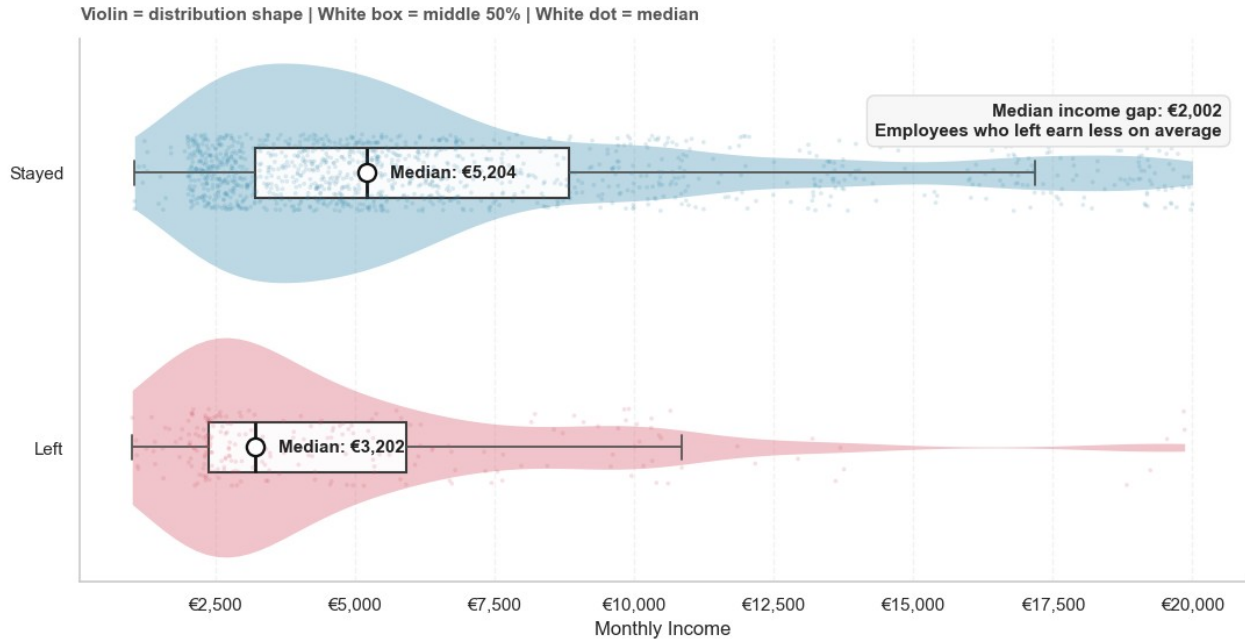


Figure 15. Monthly Income Distribution by Attrition Status.

Question answered	Do leavers earn less?
Calculation / method	Plot MonthlyIncome distribution split by Attrition. Median and distribution shape are compared.
How to read it	Leavers have lower median income; stayed employees include more high-income observations.

6. Attrition Rate by Age Band

Attrition Rate by Age Band

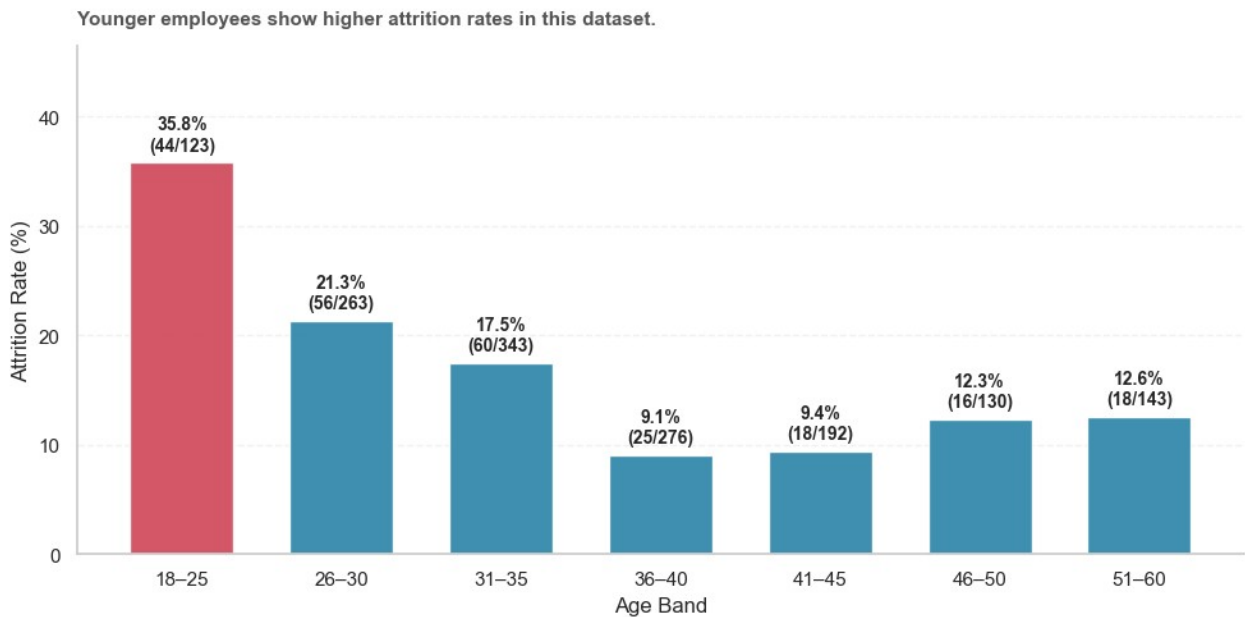


Figure 16. Attrition Rate by Age Band.

Question answered	Which age groups leave more?
Calculation / method	Create age bands with pd.cut, groupby band, calculate attrition rate.
How to read it	Young employees, especially 18-25, have the highest attrition.

7. Attrition Rate by Years at Company

Attrition Rate by Years at Company

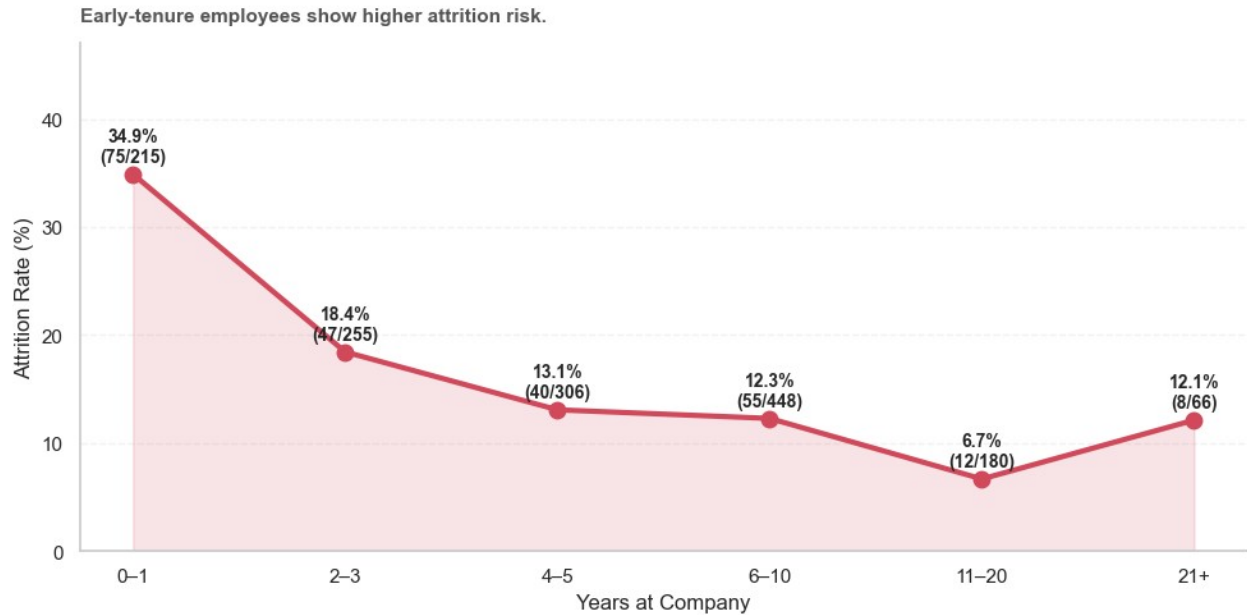


Figure 17. Attrition Rate by Years at Company.

Question answered	Does tenure matter?
Calculation / method	Create tenure bands; groupby YearsAtCompanyBand; calculate attrition rate.
How to read it	Early-tenure employees have the highest risk.

8. Attrition Rate by Job Satisfaction

Attrition Rate by Job Satisfaction

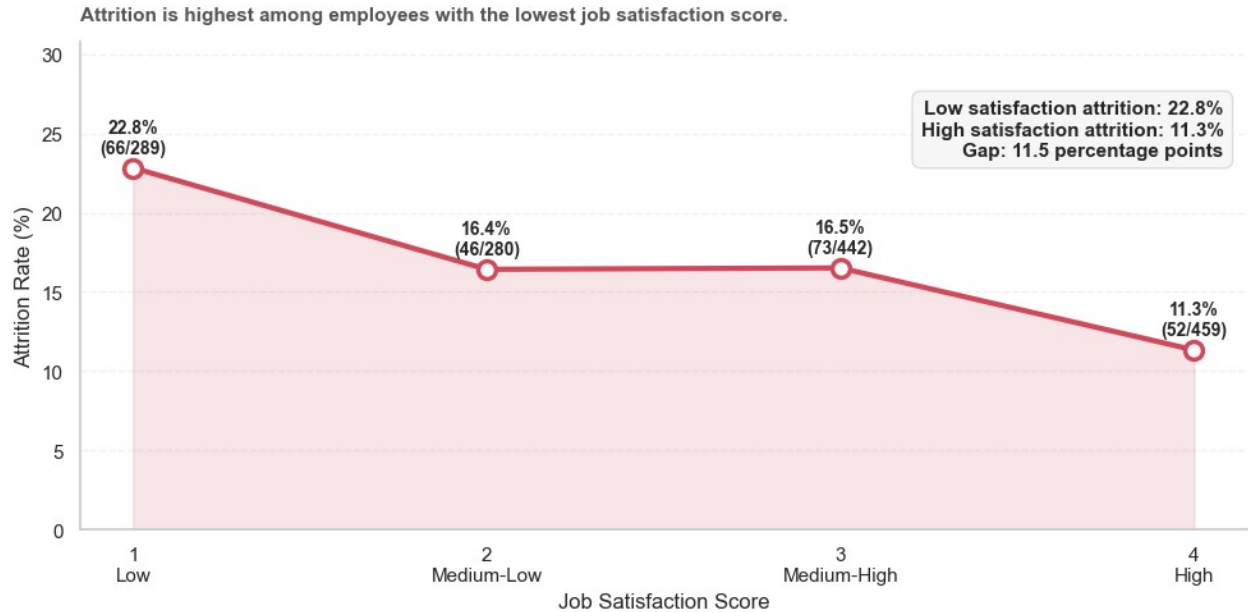


Figure 18. Attrition Rate by Job Satisfaction.

Question answered	Does satisfaction relate to attrition?
Calculation / method	groupby JobSatisfaction; calculate attrition rate per score.
How to read it	Lowest satisfaction has higher attrition; highest satisfaction has lower attrition.

9. Attrition Rate by Work-Life Balance

Attrition Rate by Work-Life Balance

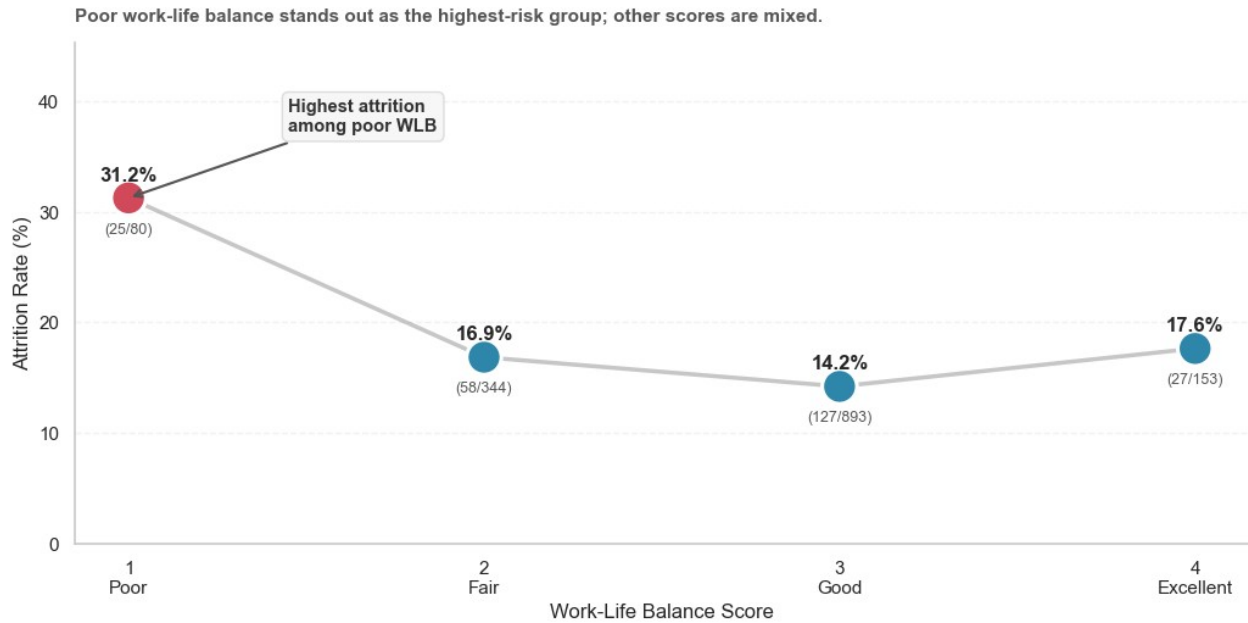


Figure 19. Attrition Rate by Work-Life Balance.

Question answered	Does work-life balance matter?
Calculation / method	groupby WorkLifeBalance; calculate attrition rate and plot as a line/dot chart.
How to read it	Poor work-life balance stands out as high-risk.

10. Attrition Rate by Environment Satisfaction

Attrition Rate by Environment Satisfaction

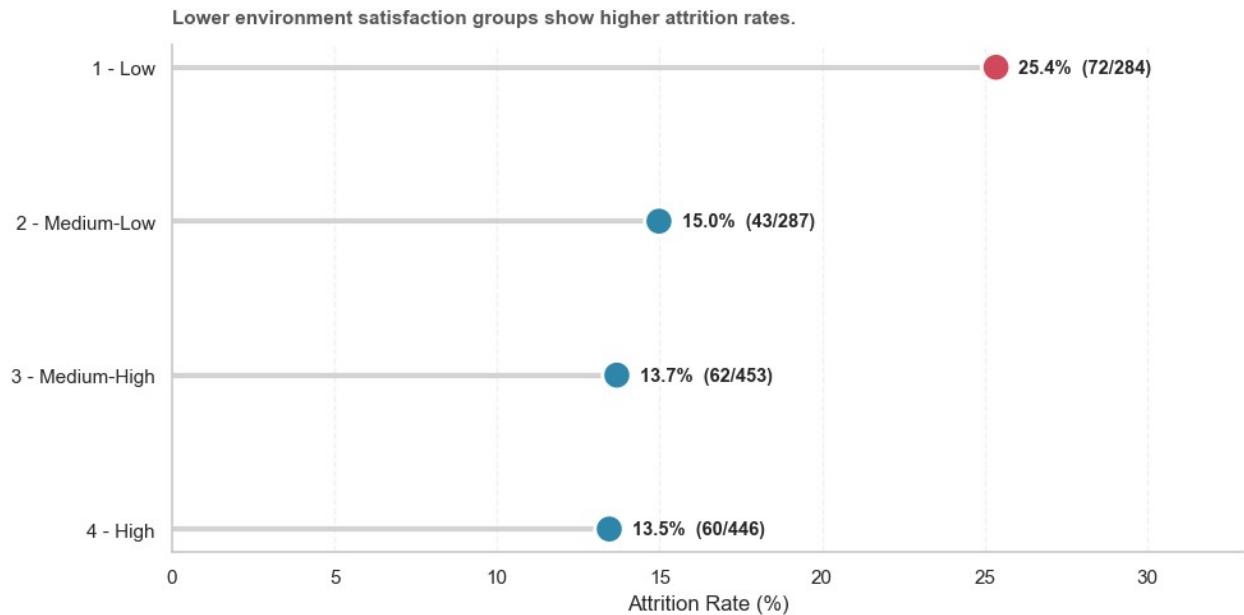


Figure 20. Attrition Rate by Environment Satisfaction.

Question answered	Does work environment matter?
Calculation / method	groupby EnvironmentSatisfaction; calculate attrition rate.

How to read it

Low environment satisfaction has the highest attrition.

11. Top Attrition Risk Segments

Top Attrition Risk Segments

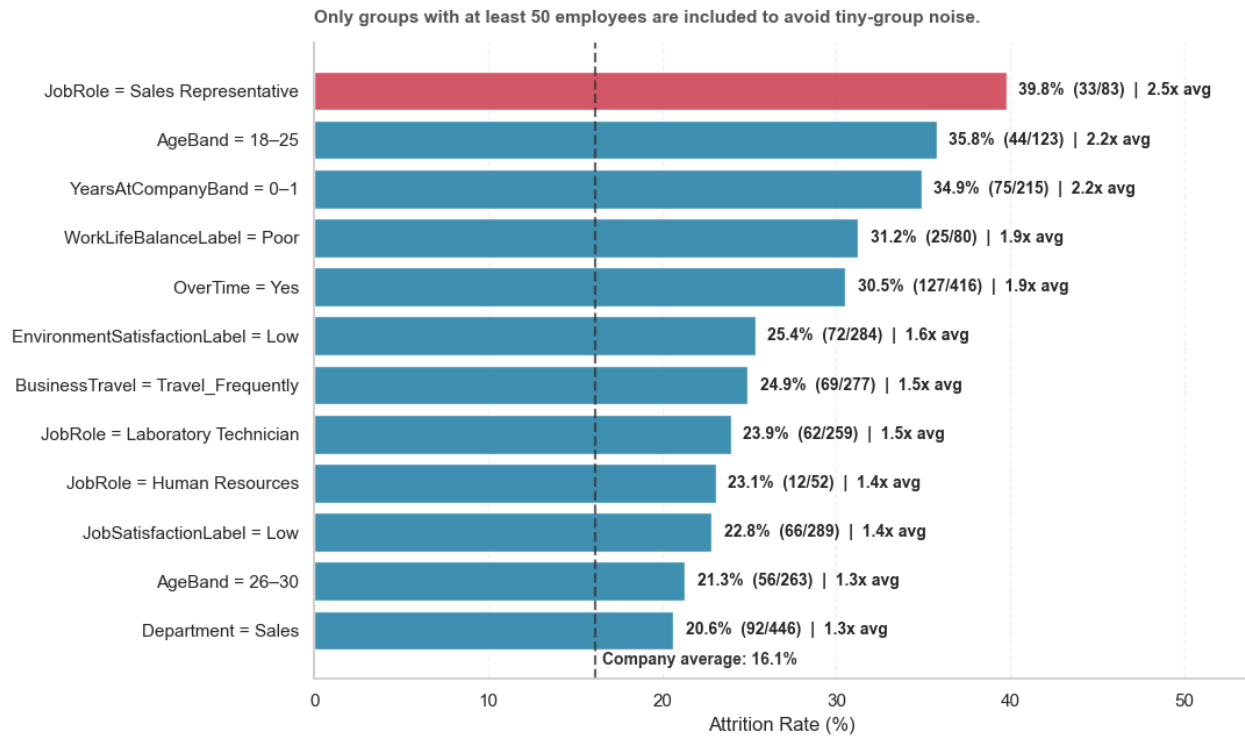


Figure 21. Top Attrition Risk Segments.

Question answered	Which groups are most risky overall?
Calculation / method	Combine selected categorical and banded features; filter groups with enough employees; calculate rate and compare to average.
How to read it	Sales Representative, young employees, early tenure, poor WLB, overtime, and low environment satisfaction stand out.

12. Numeric Feature Ranges Before Scaling

Numeric Feature Ranges Before Scaling

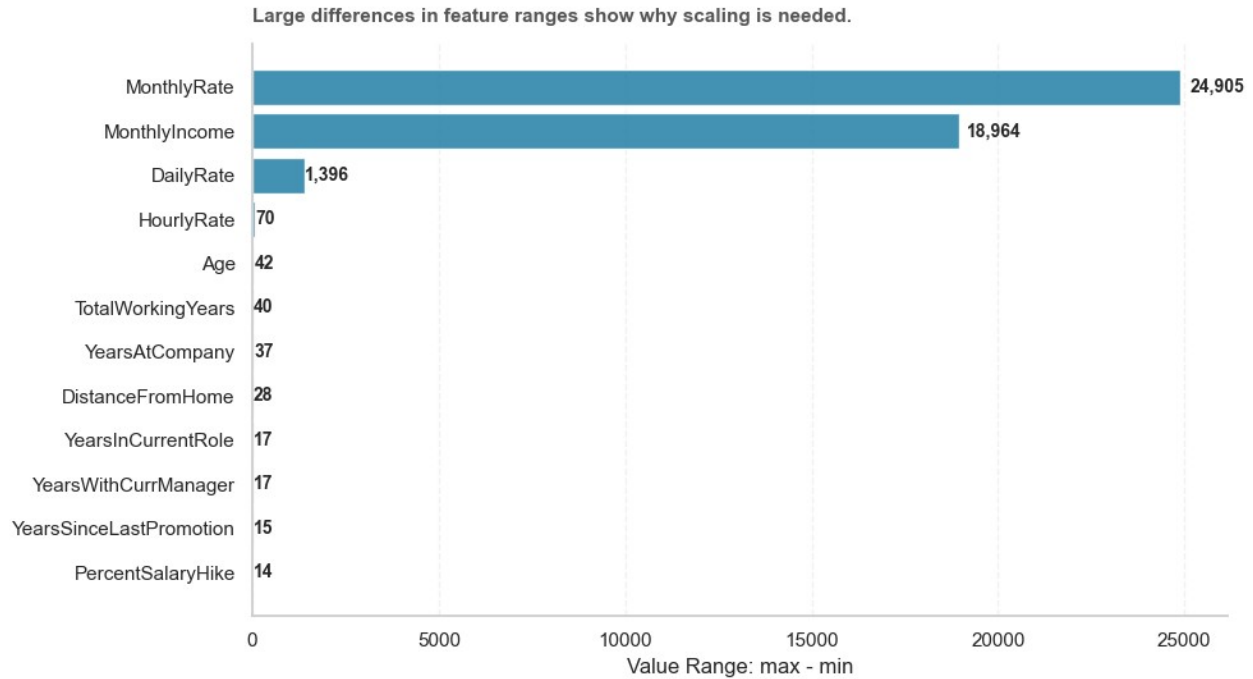


Figure 22. Numeric Feature Ranges Before Scaling.

Question answered	Why scaling is needed?
Calculation / method	For numeric columns, compute max - min and sort largest ranges.
How to read it	MonthlyRate and MonthlyIncome are much larger than satisfaction scores, so scaling helps some models.

13. Confusion Matrix

Confusion Matrix — Baseline Logistic Regression

Shows correct predictions and model errors on the test set.



Figure 23. Confusion Matrix.

Question answered	What mistakes did the model make?
Calculation / method	Compare y_{test} with predicted labels; count TP, TN, FP, FN.
How to read it	Shows false alarms and missed leavers directly.

14. Precision-Recall Curve

Precision-Recall Curve — Baseline Model

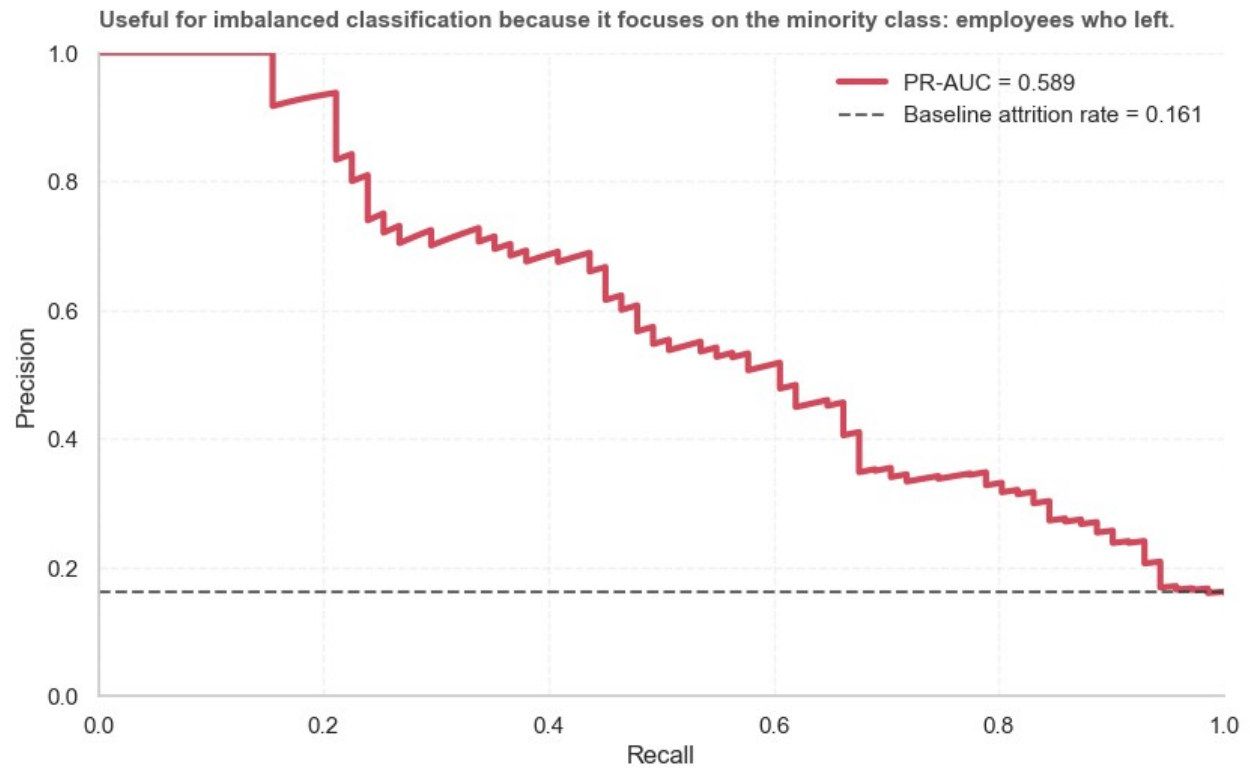


Figure 24. Precision-Recall Curve.

Question answered	How does precision trade off with recall?
Calculation / method	Use predicted probabilities over many thresholds.
How to read it	Useful for imbalanced data because it focuses on the Left class.

15. ROC Curve

ROC Curve — Baseline Model

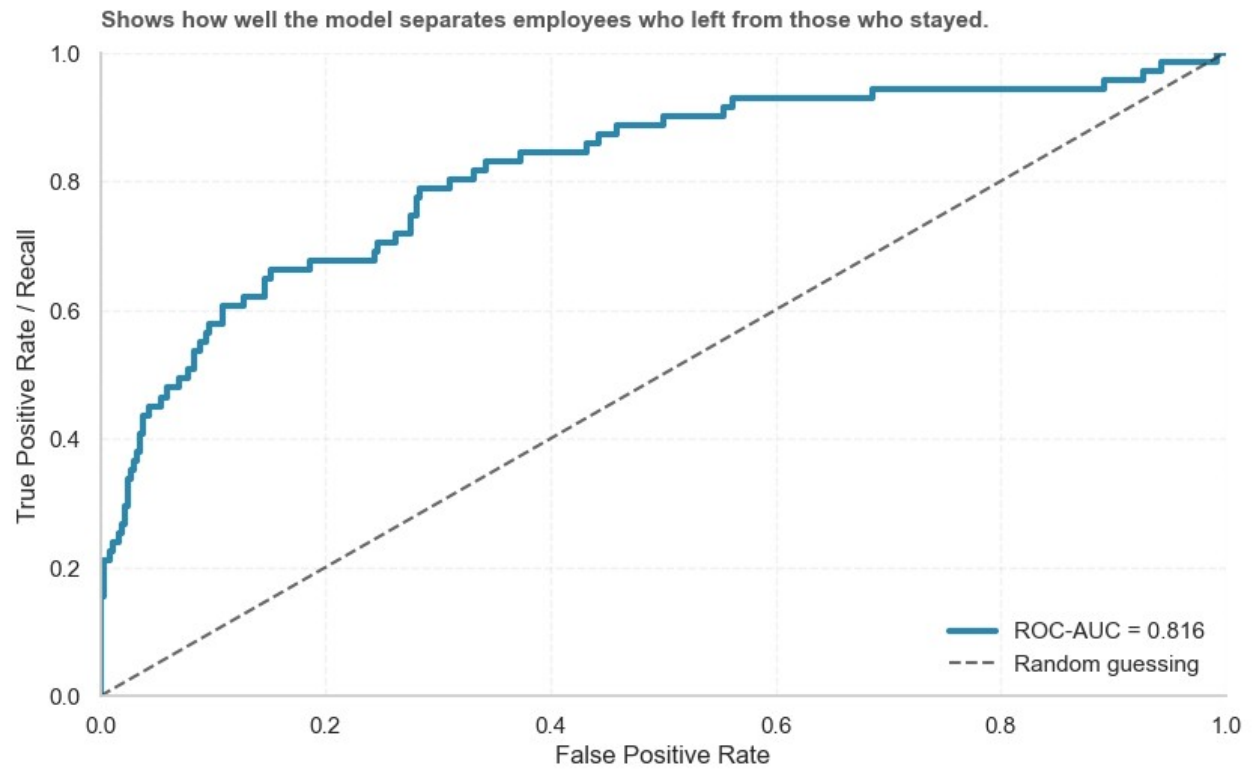


Figure 25. ROC Curve.

Question answered	Can the model rank leavers above stayers?
Calculation / method	Use predicted probabilities to calculate TPR vs FPR across thresholds.
How to read it	Curve above diagonal means better than random guessing.

16. Logistic Regression Coefficients

Top Logistic Regression Coefficients

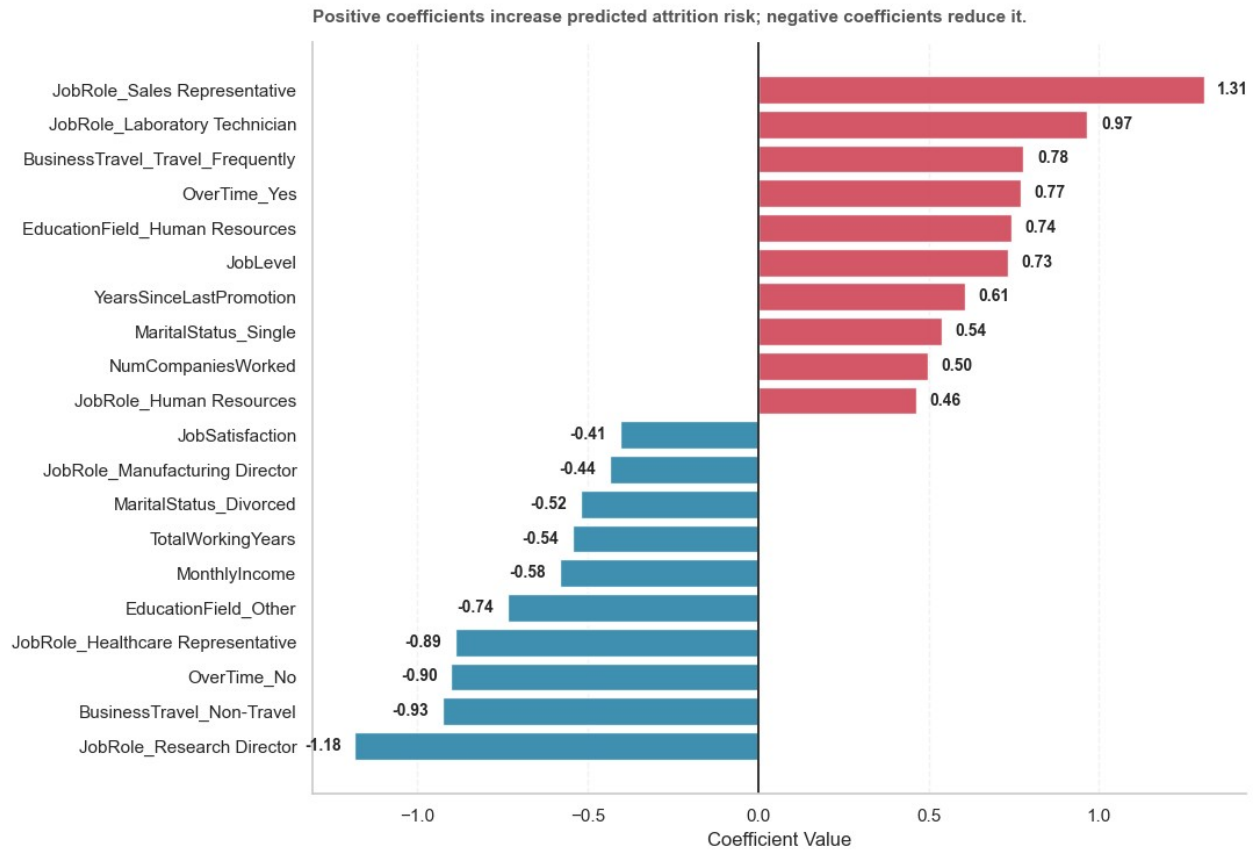


Figure 26. Logistic Regression Coefficients.

Question answered	Which features push risk up or down?
Calculation / method	Extract <code>model.coef_</code> and map coefficients to feature names.
How to read it	Positive coefficients increase predicted attrition risk; negative reduce it.

17. Scaler Comparison

Scaler Comparison for Logistic Regression

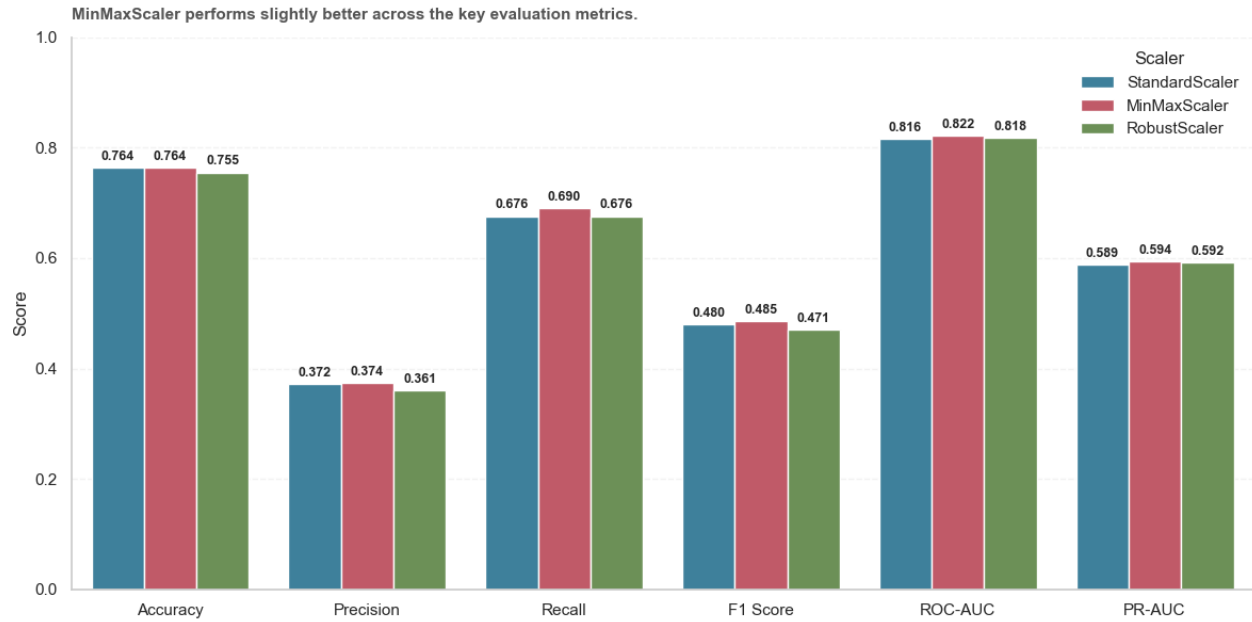


Figure 27. Scaler Comparison.

Question answered	Which scaler helped Logistic Regression most?
Calculation / method	Train same model with different scalers and compare metrics.
How to read it	MinMaxScaler performed slightly better.

18. Final Model Comparison

Final Model Comparison

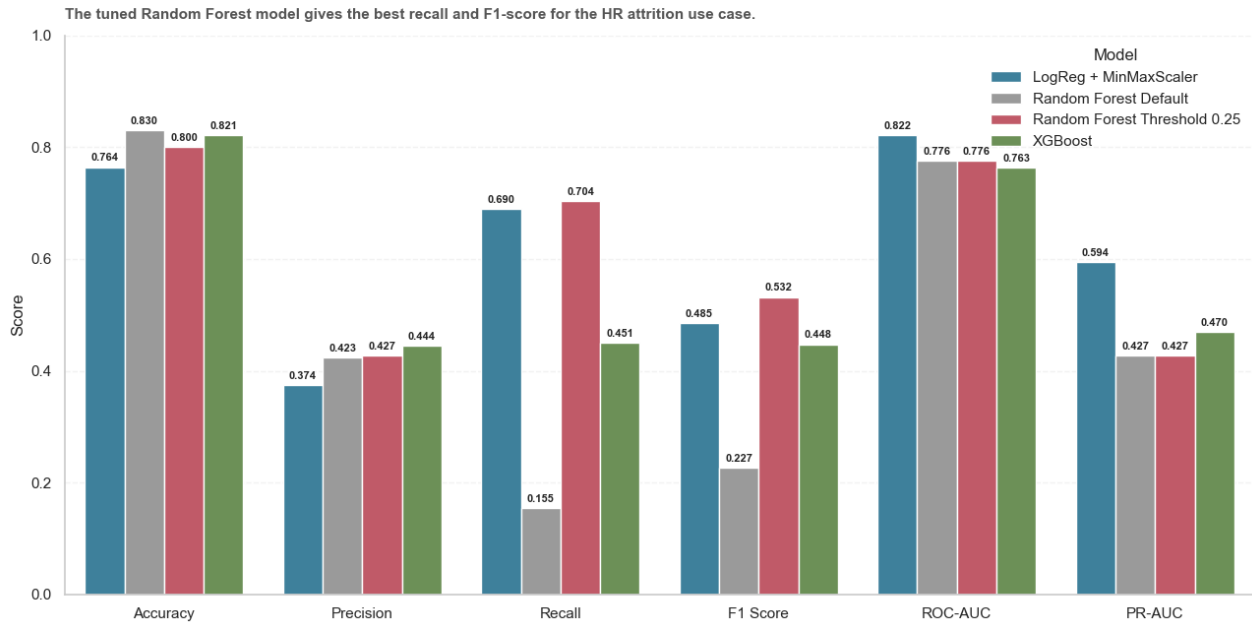


Figure 28. Final Model Comparison.

Question answered	Which final model should we select?
Calculation / method	Compare strongest candidate models across metrics.
How to read it	Random Forest threshold 0.25 wins for recall and F1-score.

10. Glossary: Difficult Terms in Plain Language

Term	Plain meaning
Feature	An input column used by the model. Example: Age, JobRole, MonthlyIncome.
Target	The answer column we want to predict. In this project: Attrition.
Classification	A prediction problem where the output is a category, such as Stayed or Left.
Pipeline	A chained workflow that keeps preprocessing and model training together.
Imputation	Filling missing values. Median for numbers, most frequent for categories.
Encoding	Turning text categories into numbers.
Scaling	Changing numeric ranges so models can compare features more fairly.
Interaction feature	A new feature created by combining two features. Example: JobSatisfaction × YearsAtCompany.
Data leakage	When information from test/future data accidentally enters training.
Threshold	The probability cutoff used to convert risk probability into a class label.
False positive	Model predicts Left, but employee stayed. A false alarm.
False negative	Model predicts Stayed, but employee left. A missed leaver.
Recall	Of all real leavers, how many did we catch?
Precision	When the model flags a leaver, how often is it correct?
F1-score	A balance score between precision and recall.
ROC-AUC	How well the model ranks leavers above stayers overall.
PR-AUC	How well the model handles the rare positive class across thresholds.

References and Sources

These sources were used to ground the technical definitions and terminology in this manual. The project-specific numbers and charts come from the IBM HR Analytics notebook built during the session.

[1] **scikit-learn StandardScaler documentation:**

<https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>

[2] **scikit-learn MinMaxScaler documentation:**

<https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html>

[3] **scikit-learn RobustScaler documentation:**

<https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.RobustScaler.html>

[4] **scikit-learn Model evaluation documentation:** https://scikit-learn.org/stable/modules/model_evaluation.html

[5] **scikit-learn Pipeline documentation:** <https://scikit-learn.org/stable/modules/generated/sklearn.pipeline.Pipeline.html>

[6] **scikit-learn ColumnTransformer documentation:**

<https://scikit-learn.org/stable/modules/generated/sklearn.compose.ColumnTransformer.html>

[7] **scikit-learn LogisticRegression documentation:**

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

[8] **scikit-learn RandomForestClassifier documentation:**

<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>

[9] **XGBoost XGBClassifier Python API documentation:**

https://xgboost.readthedocs.io/en/stable/python/python_api.html#xgboost.XGBClassifier

[10] **Category Encoders TargetEncoder documentation:** https://contrib.scikit-learn.org/category_encoders/targetencoder.html